

Zihao Wang

404-952-0911 | ziw140@ucsd.edu | gdfwj.github.io | github.com/gdfwj

EDUCATION

University of California, Irvine

Ph.D. student in Computer Science

Irvine, CA

Sep. 2026 – Present

University of California, San Diego

Master of Science in Computer Science

La Jolla, CA

Sep. 2024 – Jun. 2026

- Cumulative GPA: 3.91/4.00

- Key courses: Systems for LLMs and AI Agents, Differentiable Programming (A+), ML: Learning Algorithms (A+)

Beihang University

Bachelor of Engineering in Computer Science and Technology

Beijing, China

Sep. 2020 – Jun. 2024

- GPA: 3.92/4.0 (WES)

RESEARCH INTERESTS

Distributed systems for edge LLM serving; resource-efficient inference; sensor-grounded reasoning; privacy-aware and connectivity-resilient smart spaces.

SCHOLARSHIPS & AWARDS

- **2021, 2022, 2023** Academic Merit Scholarship (Awarded to undergraduate students for outstanding GPA throughout the school year)
- **2022, 2023** Academic Competition Scholarship (Awarded to undergraduate students who achieved outstanding rankings in national or university-level discipline competitions)

PUBLICATIONS & ACCEPTED PAPERS

1. Ye Tian*, **Zihao Wang***, Onat Gungor, Xiaoran Fan, Tajana Rosing. “LifeAgentBench: Benchmarking LLMs for Long-Horizon, Cross-Dimensional Lifestyle Health Reasoning.” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, main conference, 2026, accepted. *Equal contribution
2. Ye Tian, Xiaoyuan Ren, **Zihao Wang**, Onat Gungor, Xiaofan Yu, Tajana Rosing. “DailyLLM: Context-Aware Activity Log Generation Using Multi-Modal Sensors and LLMs.” *IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS)*, 2025, pp. 372–380

PREPRINTS & MANUSCRIPTS

1. Ye Tian, Jingyi Zhang, **Zihao Wang**, Xiaoyuan Ren, Xiaofan Yu, Onat Gungor, Tajana Rosing. “KLDive: Fine-Grained 3D Scene Reasoning for Autonomous Driving based on Knowledge Graph.” *arXiv:2603.21029*; under review at AAAI, 2026
2. **Zihao Wang**, Jing Zhao, Xuotong Ding, Hui Zhang. “Double-Flow GAN model for the reconstruction of perceived faces from brain activities.” *arXiv:2312.07478*, 2023

RESEARCH EXPERIENCE

Distributed LLM Serving in Pure-Edge Smart Spaces

Sep. 2026–Present

Doctoral Research, Advisors: Nalini Venkatasubramanian

University of California, Irvine

- Defining research questions for distributed LLM serving across local edge devices without relying on cloud inference.
- Studying how LLM services can incorporate sensor and contextual data from the surrounding physical space to support locally grounded reasoning.
- Targeting privacy-sensitive and intermittently connected smart spaces, with potential applications in homes, care environments, and remote monitoring sites.

LifeAgentBench: Lifestyle Reasoning Benchmark

Jul. 2025–Sep. 2025

Research Assistant, Advisors: Ye Tian and Tajana Rosing

System Energy Efficiency Lab, University of California, San Diego

- Built **LifeAgentBench**, a large-scale benchmark (22,573 questions) spanning single-domain retrieval to long-horizon, cross-dimensional reasoning over lifestyle health records in a structured database.
- Built automated evaluation pipelines for context-based reasoning and schema-to-SQL execution; the latest benchmark study evaluates 13 LLMs.
- Implemented **LifeAgent**, a training-free, tool-augmented reasoning baseline using question decomposition, iterative retrieval, and deterministic computation.
- The paper reports 40.16% average accuracy on challenging subsets with Qwen2.5-7B, versus 7.74% for Context Prompting and 9.43% for Database-augmented Prompting.
- Released code on GitHub and the manuscript on arXiv.

DailyLLM: Multimodal Activity Log Generation

Jan. 2025–Apr. 2025

Research Assistant, Advisors: Ye Tian and Tajana Rosing

System Energy Efficiency Lab, University of California, San Diego

- Developed **DailyLLM**, a lightweight LLM-based system for context-aware activity log generation and summarization from multimodal smartphone and smartwatch sensors.
- Organized IMU-based datasets for daily activity understanding, including data preprocessing, augmentation and data construction for model training.
- Built multi-task LoRA fine-tuning and evaluation pipelines to balance model capability and system efficiency.
- Deployed and evaluated the system on Raspberry Pi; resolved a quantized-kernel compatibility issue by selecting a supported quantization bit-width. The paper reports four minutes to summarize a two-hour activity window on Raspberry Pi 5.
- Released code, data, and a project website (GitHub); paper accepted to **IEEE MASS 2025**.

Reconstruction of perceived face images from fMRI

Apr. 2023–Dec. 2023

Research Assistant, Advisors: Hui Zhang

Medical-Engineering Innovation Research Institute

- Developed a GAN-based pipeline to reconstruct face stimulus images from fMRI signals.
- Proposed a pretraining strategy on CelebA with VGGFace feature conditioning to mitigate limited paired fMRI-image data.
- Designed a double-flow discriminator to improve reconstruction fidelity.
- Manuscript released on arXiv.